

Instrumentation Lab Session 1

Contents

1. Introduction.....	1
1.1 How to log onto your lab workstation PC	2
2. Digitisation, Aliasing and Filters	3
2.1 Objectives	3
2.2 Introduction.....	3
2.3 Experiment 1 – Proper Sampling.....	3
2.4 Experiment 2 – Aliasing	4
2.5 Experiment 3 – Apparent Frequency of the Aliased Components	4
2.6 Experiment 4 – Complex Waveforms and Aliasing	4
2.7 Experiment 5 – Corruption of the Properly Sampled Components	5
2.8 Experiment 6 – Sampling an Analogue Waveform; The Nyquist Criterion.....	5
2.9 Experiment 7 – Building and Characterising the Anti-Alias Filter	7
2.10 Experiment 8 – Applying the Anti-Alias Filter	10
3. If you have time.....	11

1. Introduction

There are a couple of objectives for this first session in the lab

- To familiarise yourself with the LabVIEW working environment, the ELVIS electronics prototyping boards, and the National Instruments “Software Front Panel” (SFP) instruments
- To do a practical investigation into the sampling, digitising and aliasing topics which we have covered in lectures and in the handout

There won’t be much electronics to work on in this first session, but you’ll build a simple RC filter.

Work on sensors, signals, op-amps and transfer functions will come in later sessions.

You should aim to complete as much of this script as possible during this lab session.

Therefore, please keep an eye on your progress and if you get stuck ask a demonstrator.

Note that since you are working in pairs please ensure that you both understand the material before proceeding through the script, and that you both have an adequate share of working with both the software and hardware aspects of the lab. For record-keeping, it is

convenient to take screen-shots which you can store temporarily in a folder on the desktop. However, since you will probably use a different workstation each lab, please copy these files off onto a memory-stick at the end of each lab session. How to log onto your lab workstation PC

User name: your Imperial user name

Password: your Imperial password

Note: Case Sensitive!

Hopefully, you have already read the handout on sampling and digitisation. This will give you background information needed for this lab script.

How to power up NI ELVIS:

After PC boot up:

- Make sure NI ELVIS is connected via USB to the PC (USB port is on the rear panel of ELVIS Bench-top Workstation)
- Power up ELVIS using the switch on the back panel, not the switch on the top, which is for powering up the prototyping board.
- Windows will recognise a new USB device. If the ***New Data Acquisition Device*** windows pop up, select ***Take No Action*** and click ***OK***. If you get a pop-up asking you if you want to run ***ELVISmx Instrument Launcher*** click ***OK***

How to run "ELVISmx Instrument Launcher":

- Navigate to Start > Programs > National Instruments > Ni ELVISmx > NI ELVIS mx Instrument Launcher

If you want to record screenshots from LabVIEW :

Either print directly to the printer or you can print HTML format to a file (images as PNG files). Make sure if you save images from LabVIEW that you rename them and move them to a new location, as LabVIEW has a nasty habit of overwriting image files! If in doubt, ask a demonstrator.

1.1 Shutdown at the end of the lab session

If you've stored any data files or plots on the PC then take them with you on a memory stick as you might not be using the same PC for the next lab session, unless you save it into your account (not locally).

At the end of the session, please shut down the PC by pushing the on/off button on the front of the base unit (**don't** log-off first).

Switch off the ELVIS unit and remove the breadboard from the top. You may leave the circuit in-place. Put it in the box provided and record the box number

Store it away in the cupboard (ask your demonstrator where) ready to use again next time.

2. Digitisation, Aliasing and Filters

2.1 Objectives

Investigate

1. Sampled and Digitised signals
2. The Sampling Theorem and The Nyquist Frequency
3. Aliasing
4. RC Filter Frequency response and use as anti-aliasing filter

2.2 Introduction

The Nyquist criterion states

The Sampling Rate should be at least twice the highest frequency in the analogue signal

If you are sampling a sine wave signal, this is easy to achieve since the signal contains only a single frequency component. However, an arbitrary waveform such as a square wave contains many higher frequency Fourier components. We will see this in the frequency domain plots.

Aliasing is the appearance of 'phantom' frequencies when a signal is not sampled at a high enough rate. In films (which are normally sampled at 24 frames per second) we may see the wheels of cars, stand still, or even appear to rotate backwards. This is aliasing. If the sampling of the camera is precisely synchronized with the rotation of the wheel, it will appear to stand still.

In Experiments 1 to 5 we will use a simulated digitiser i.e. a piece of software that takes periodic samples from a mathematically defined waveform. In experiments 6 to 9 we will make use of a real digitiser provided by the NI ELVIS to sample an analogue waveform, also generated by NI ELVIS.

The two programs we will use, ***Aliasing.llb*** and ***FFTanalyser.vi*** are called "Virtual Instruments". They have been developed using LabVIEW and coded in G, a high-level graphical programming language.

2.3 Experiment 1 – Proper Sampling

- Run the program ***Aliasing.llb*** located on your PC desktop; when the "LLB Manager" opens, double click on the top entry ***Aliasing.vi*** to run it.
- On the ***Aliasing.vi*** front panel, select Sine Waveform and set

Signal Frequency = 450 Hz
Sampling Rate = 1000 Samples/s (S/s).

Note that the sampled (yellow) waveform is rather 'jagged' due to the relatively few samples per cycle. The display plots a yellow line between samples. However, even with these few

samples we can still make a correct measurement of the signal as can be seen in the frequency domain graph containing only one tone at 450 Hz. The frequency domain plot is calculated in 'real-time' by LabVIEW using a Fast Fourier Transform (FFT) algorithm.

- Click the mouse to the right of the last digit in the box below the Signal Frequency Slider. You can now change the last digit of the Signal Frequency, using the Up/Down keys on your keyboard. Hold down the Down Arrow key, and note how the points of the waveform that are sampled change quite dramatically.

2.4 Experiment 2 – Aliasing

- Now, increase the frequency by holding down the Up Arrow key on your keyboard, as in Experiment 1.
- Notice that when you pass 500 Hz, three things happen: the sampled waveform has only two samples per period, and looks almost flat, and the Frequency Spectrum now shows the peak moving backwards (decreasing in frequency) as you increase the frequency. This is Aliasing, which is why the Aliasing indicator comes on (Note that because this is a simulation the software *knows* when aliasing will occur; real life is not so simple, as we will see)
- Keep increasing the Frequency until you get close to 1000 Hz. As you slowly approach 1000 Hz, note that the Tone on the Frequency Spectrum finally disappears at 0 (DC) and then re-emerges as you increase the frequency, this time running the right direction, but with measured value that is 1000 Hz too low on the Frequency Spectrum.

2.5 Experiment 3 – Apparent Frequency of the Aliased Components

- Set the Signal Frequency to just below 1500 Hz. Notice that the sampled waveform repeats only 1/3 times as often as the original!

With aliasing, you cannot tell the aliased components from the real ones

- Recalling the handout on Digital Signals, ensure that you understand how aliasing occurs at the theoretical level and also that the aliased frequencies observed in these experiments are the frequencies you would expect to see.

2.6 Experiment 4 – Complex Waveforms and Aliasing

- Repeat the above experiments using a Square Waveform. Since the square wave contains many higher frequencies (harmonics) this will alias dramatically, and you may see various portions of the spectrum both moving up and down at the same time as you change the Frequency of the Square Wave. This is a dramatic illustration of the mess that aliasing can cause

Note: The Fourier handout gives the Fourier series for the square wave, which may be useful at this point.

2.7 Experiment 5 – Corruption of the Properly Sampled Components

- With a Square wave, set

Signal Frequency to 250 Hz

Sampling Rate to 5000 Hz.

The Spectrum looks nice and free from Aliasing, however we know that this cannot be the case.

- Increase the Signal Frequency to 260 Hz in steps of 1 Hz and notice all the aliasing components that show up. Notice also how the harmonic levels of the spectrum drop.

This is because the signal at 250 Hz also was aliased, but the frequencies of the aliasing components fell on top of the correctly-sampled frequency components in the square wave, thus making the aliasing invisible, and giving erroneous measurements for the values of the properly-sampled components.

Aliased frequency components can corrupt measurements of real frequency components

Make sure you understand what's happening here before you move on. Discuss with your demonstrator.

2.8 Experiment 6 – Sampling an Analogue Waveform; The Nyquist Criterion

In this experiment we will use the ELVIS digitiser to acquire a real analogue signal. Once digitised and transferred onto the PC memory, the signal will be processed using a FFT algorithm. The combination of a digitiser and an FFT processor is usually referred to as a 'FFT analyser'.

The FFT analyser has been implemented using LabVIEW. The program **FFTAnalyser.vi** makes use of ELVIS to digitise and transfer into the PC memory (via USB port) the signal under test. It then applies a FFT algorithm to transform the time-domain sampled signal into the frequency domain. The results are displayed graphically.

The analogue signal (test signal) will be generated using ELVIS's digital-to-analogue converter (DAC).

FGEN set up

- From the **NI ELVISmx Instrument Launcher** open the **Function Generator (FGEN)**.
- Use the context help accessible via the **Help** button to learn more about **FGEN**.

- On the **FGEN** front panel, select:

Sine Waveform

Frequency = 100 Hz

Amplitude = 1 V_{pp}

DC Offset = 0 V

Signal Route = FGEN BNC

Note: this is the name of the physical ELVIS

connection at which the generated signal will be available

The other settings are irrelevant for this set of experiments.

- Click the Run button to start generating the waveform.

Tektronix oscilloscope set up

- Set the Tektronix so that you can see the waveform being generated.

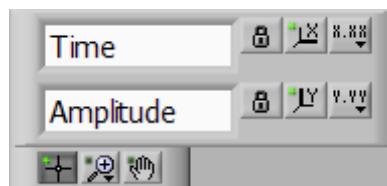
FFT analyzer set up

- On the ELVIS side panel, connect FGEN BNC to CH0 BNC (the input of the ELVIS digitiser)
- Launch the FFT analyser by opening the program **FFTAnalyser.vi** located at on your PC's desktop.
- On **FFTAnalyser.vi** front panel, set:

Sampling Rate = 1000 S/s

Number of samples = 100 S

Clicking the button **Run**  the programme will execute once (i.e. will acquire and process the signal); the button **Run Continuously**  will instead execute the program continuously. We will use the latter.



Use **Cursor and Graph Controls** on the right of the Acquired Waveform graph to remove the X auto scale and then zoom in the acquired signal. Familiarise yourself with the different functions offered by the controls.

Notice that the acquired signal looks similar to what's on the Tektronix display and the FFT shows no obvious presence of aliasing: only the tone at 100Hz is present.

- Increase the frequency of the test signal to 330Hz and then to 480Hz.

Notice that:

1. The acquired waveform is no longer a pure sine wave

2. The acquired signal looks different from what's on the Tektronix display (adjust the settings of the oscilloscope if necessary)
 3. The FFT main tone shows that the frequency of the acquired signal is as expected i.e. 330Hz and 480Hz.
- Can you explain what's happening? Is this 'proper sampling', as per the definition in the handout?
 - Now increase the Sample Rate to 10000 S/s.

The acquired waveform now looks like the one on the Tektronix's display.

What you were observing was the effect of artefacts on the acquired signal due to particular combinations of sample rate and signal frequency.

The Tektronix is a digital oscilloscope and its digitizer works pretty much like NI ELVIS's. The reason why the Tektronix always seems to display the "correct" waveform is because its samples rate is very high (2.5 GS/s). The same artefacts will appear in the Tektronix if you try to acquire a very fast sine wave.

Remarks

Regarding the sample rate we can conclude that:

1. It must be at least twice the highest frequency under measurement to satisfy the Nyquist criteria and avoid aliasing;
2. Digitised waveforms are always an "approximation" of the analogue waveform. Having a high sample rate helps to increase the fidelity of the acquired signal.

2.9 Experiment 7 – Building and Characterising the Anti-Alias Filter

- Before starting, switch-off the prototyping board and stop FGEN (if running).

For proper sampling we must try not to violate the Nyquist criterion.

We can do this by filtering the signal to be measured. This is done using a low-pass filter which attempts to remove frequency components higher than the Nyquist. It is normally called an **anti-aliasing filter**.

Frequency domain lab equipment (spectrum analyser, dynamic signal analyser etc.) comes equipped with a built-in anti-aliasing filter, since the main function is to accurately represent the frequency content of the signal. The Oscilloscope usually has no anti-aliasing filters since its job is to measure waveforms. The responsibility to avoid aliasing is left to the user.

In this experiment we will build a low pass filter which you will then be used as an anti-aliasing filter.

Recalling the material from the lectures, for an RC low-pass filter the cut-off frequency is given by

$$f_{cutoff} = \frac{1}{2\pi RC}$$

Where:

f_{cutoff} = frequency at which $\frac{V_{out}}{V_{in}} = \frac{1}{\sqrt{2}}$ (also referred to as the *3dB point*)

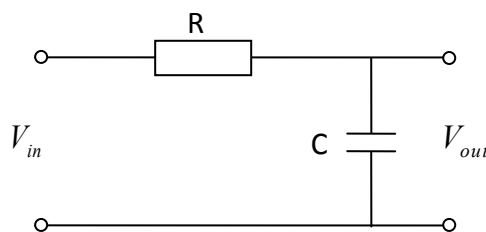


Figure 1, Low-Pass RC Filter

Clearly, f_{cutoff} is determined by our chosen Sampling Rate. Assuming we are going to sample at 10000 S/s, if we want to satisfy the Nyquist criteria we need a cut-off frequency of 5 kHz.

- Implement on the NI ELVIS prototyping board an RC low-pass filter with $f_{cutoff} = 5$ kHz. What values of R and C do you choose?

Hint: choose a resistor from one of the standard values in the range 5kΩ to 10kΩ, such that a standard value capacitor gives a reasonably close match to the desired cut-off frequency.

Testing the Filter

The most important piece of information regarding a filter is its frequency response. This is defined in terms of Gain (V_{out} / V_{in}) and Phase (Phase difference between V_{out} and V_{in}) versus frequency.

The frequency response fully defines a filter's characteristics.

The frequency response can be measured using a Bode Analyser. This instrument will step through a range of frequencies and measure the Gain and Phase of the device under test (not necessarily a filter) at each step.

- Launch the Bode Analyser using the button **Bode** on the ELVISmx Instrument Launcher.
- Read section **Bode Analyzer Soft Front Panel Controls** in the context help to learn about the instrument functions and controls available.

Connections:

You need to provide the Bode Analyser with the filter's input and output signals. The filter's input signal comes from FGEN (+) and GROUND (-) on the prototyping board.

- Read section **Taking a Measurement** in the context help to learn about connections. Concentrate on step 2, 3 and 4. Figure 2 shows where the relevant terminals are.

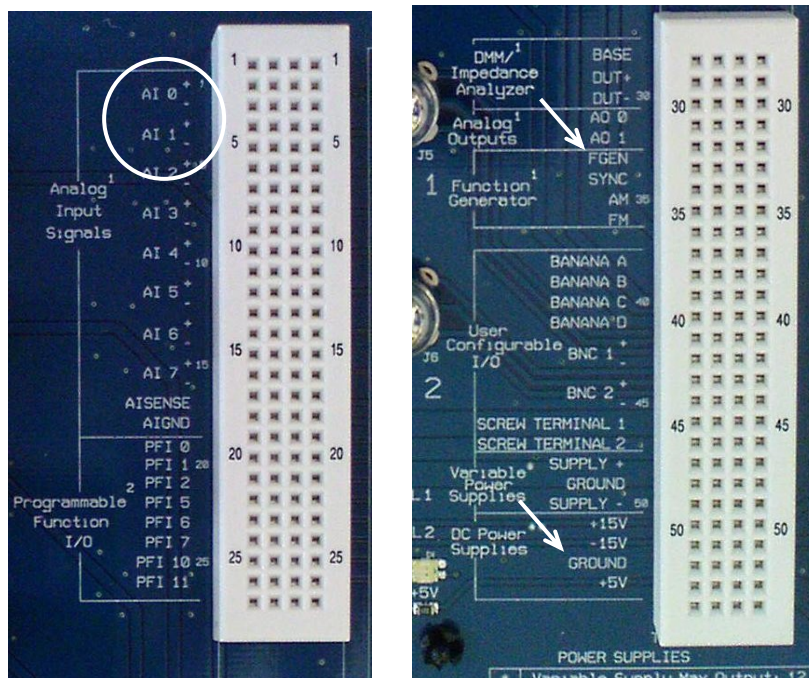


Figure 2, Connections for the Bode Analyser

After the connections are in place:

- On the Bode Analyzer, set:

Start and Stop frequencies to cover the range of interest;

Frequency Step so that the measurement doesn't take too long and allows you to carry out the measurements requested below;

Amplitude = $2 V_{pp}$

- Switch on the prototyping board.
- Click **Run** on the Bode Analyzer to start measuring the frequency response.
- Use the cursor to measure:
 1. The cut-off frequency of the filter and determine its actual bandwidth
 2. The filter's roll-off in dB/decade

Do these results match your predictions?

2.10 Experiment 8 – Applying the Anti-Alias Filter

Now that you have characterised your filter, you can use it as an anti-aliasing filter.

- Switch off the prototyping board before making the following connections

Connections:

1. Connect the filter's input to BNC1+
2. Connect filter output to BNC2+
3. Connect BNC1– and BNC2– to GROUND

The filter's input and output signals are now available on the two connectors labelled **BNC1** and **BNC2** on the left-hand side of the prototyping board.

Connect BNC1 and BNC2 to CH0 and CH1 respectively.

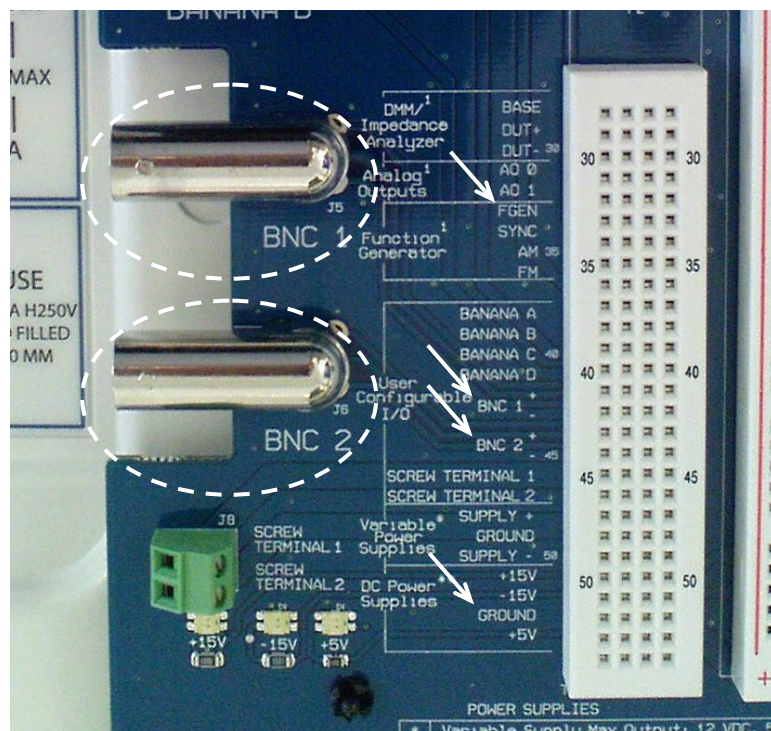


Figure 3, BNC Connections

The filter's input and output are now available on NI ELVIS' "scopeCH0" and "scopeCH1" channels respectively.

- Switch on prototyping board.
- Set FGEN and Run

Square wave

Frequency = 700 Hz

Amplitude = 1 V_{pp}

Signal Route = Prototyping board

- On FFTAnalyser.vi front panel, set:

Sampling Rate = 10000 S/s
Number of samples = 4000 S

- Let's take a look at the spectrum of the unfiltered signal. On FFTAnalyser.vi front panel, set:

Input Channel = scopeCH0

On the FFT display you should see the effect of aliasing.

- According to theory, the spectrum of a 700Hz square wave should show spectral lines at what frequencies? (for the first, second, third order harmonics)

Along with these, you will notice a myriad of other lines, most of which are higher order harmonics aliasing through.

- Now let's consider the filter's output spectrum: on FFTAnalyser.vi front panel, set:

Input Channel = scopeCH1

- What can you observe now? Does the result confirm the theory? Is aliasing still present? What happened to the amplitude of the fundamental harmonic of the input signal?

3. If you have time...

From your knowledge of the Fourier synthesis of signals you should be able to calculate the relative powers of the fundamental frequency and the first few harmonics of the square wave. Experiment a bit with various FGEN frequencies and relate your observations to the calculated values. Try this initially without the filter and then with the filter. Does the measured attenuation due to the filter match the results of the Bode analyser?

Try this also with other waveforms, such as sawtooth or triangle.

In the frequency domain, estimate the smallest and largest signals which can be measured. What is the Dynamic Range of this measurement?

Ask your demonstrator about the performance of the ELVIS system. What is the digital resolution of the ADC? (How many bits?) What is the input voltage range? Calculate the theoretical maximum quantisation error in both bits and volts. Express the quantisation noise added to the analogue signal during the sampling process.

Adapted from resources available on the National Instruments website

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<http://zone.ni.com/devzone/cda/epd/p/id/4852> - Introduction to NI ELVIS

<http://zone.ni.com/devzone/cda/epd/p/id/1267>